

FPT University HCM



Deliverable 2

Topic: Agriculture UAV

**A platform for managing and
coordinating regional precision
agricultural spraying drone
services. Copper Mekong River**

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Subject: ITA301

Instructor: *Ngo Dang Ha An*

Ho Chi Minh City, May 2026

PROJECT INFORMATION

1. **Department responsible:** This project was carried out by a group of students currently studying at FPT University Ho Chi Minh City as part of their ITA301 – Systems Analysis and Design course.

List of Project Members

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2. **Content to be implemented:** Research and analysis of a platform supporting the coordination of precision agricultural spraying drones aims to optimize UAV management, spray scheduling, task progress monitoring, and support the optimization of operational models and management of precision agricultural services.

Research project objectives:

- Optimize human resources, minimize scheduling conflicts and coordination errors.
 - Increase the efficiency of pesticide spraying operations using UAVs.
 - Monitoring task progress at a visual, operational level and enhancing the transparency of log data.
 - Contributing to the advancement of digital transformation in the field of precision agriculture in Vietnam.
3. **Execution time:** May 2026 – August 2026.

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1. Requirement Elicitation Approach

The requirements were collected using various methods to reflect the actual operating procedures of businesses providing agricultural UAV services..

Literature Review

Study the documents:

- IEEE Std 830-1998: ISRS is an IEEE standard used to guide the writing of Software Requirements Specification (SRS) documents.
- APA 7th Edition (American Psychological Association).
- Cost and Workload Assessment of Agricultural Drone Sprayer
- The Economic Effects of Unmanned Aerial Vehicles in Pesticide Application

Domain Analysis

Analyzing the business processes of companies providing drone-based pesticide spraying services.

Stakeholder Analysis

Identify the stakeholders:

- Administrator
- Manager
- Farmer
- Agronomist
- Drone Operator
- Technician

Requirement Modeling

The requirements are expressed as follows:

- Functional Requirements (FR)

- Non-functional Requirements (NFR)
- Business Rules (BR)
- Use Cases (UC)
- Requirement Traceability Matrix (RTM)

2. User Classes and Actors

Stakeholder	System Actor / User Class	Need	Main Permissions
	Administrator	A centralized system management tool, including user accounts, permissions, and system activity monitoring.	- User account management (CRUD user) - Role-based access control (assigning and revoking roles) - System configuration and data catalog - View and analyze system logs - Not involved in operational tasks
	Manager	Monitor and coordinate agricultural UAV service activities on a centralized platform.	- Review/reject service requests from Farmer - Assign tasks to Drone Operators and Agronomists - Managing flight schedules and allocating resources - Monitor the progress of task completion. - Generate operational reports
	Farmer	Request pest control services and provide the necessary information for the process.	- Create a service request - Land parcel information management - Provide information on crops and pests. - Suggest your desired medication - Monitor service status
	Agronomist	Evaluate farmers' pesticide application requirements to ensure suitability, safety, and regulatory compliance.	- Consider farmers' requests for pesticide spraying. - Checking the legality of the medicine (Ministry of Agriculture and Rural Development) - Assess suitability for the crop - Check the dosage. - Create spray prescriptions and send them to the Manager and Drone Operator.

Drone Operator	Perform pesticide spraying tasks using drones at the site.	- Receive and review assigned tasks - Download flight maps and mission data - Carry out flying and spraying operations. - Record the Operation Log and Spray Log. - Send an emergency (SOS) alert
Technician	Manage the technical aspects, maintenance, and safety of drone equipment.	- Manage drones, batteries, and charging stations. - Perform maintenance and repairs - Track device history - Approval of the pre-flight safety checklist

3. System Overview

Item	Content
Project Name (VN)	A platform for managing and coordinating precision agricultural spraying drone services.Mekong Delta region
Project Name (EN)	Agriculture UAV – Precision Spraying Management Platform
Domain	Precision Agriculture, agricultural UAV services
System Type	Management & Coordination Information System
Platform	Web Application + Mobile Application
System Scope	Managing and coordinating drone-based pesticide spraying services.
Out of Scope	This does not include automatic flight control or direct UAV control.

The purpose of the system (One-sentence System Purpose)

The system supports agricultural UAV service businesses in managing and coordinating precision spraying operations by digitizing the process of request reception, planning, task assignment, operational monitoring, and report compilation.

4. Requirement elicitation plan

The Requirement Elicitation Plan was developed to identify, collect, and validate the functional and non-functional requirements of the Agriculture UAV system. The team

applied a combination of methods to ensure that the requirements accurately reflect the actual operating procedures of the UAV spraying service, while minimizing the risk of omissions or inconsistencies in the system specification.

4.1 Role-play Interview

The team used role-play interviews to gather requirements from the perspective of key actors in the system. Instead of just asking theoretical questions, the team constructed simulated interviews based on the actual workflow of an agricultural spraying service.

Target

- Understand the responsibilities, needs, and challenges of each user group.
- Identify actual business processes and operational bottlenecks.
- Gather functional requirements (FR), business rules (BR), and non-functional criteria (NFR).

Roles involved

Actor	Interview content
Farmer	Instructions on how to register land plots, submit service requests, provide crop/pest information, track progress, make payments, and rate services.
Drone Operator	Task assignment process, pre-flight checklist, offline operations, flight/spray logbook recording.
Manager	Coordinate flight schedules, allocate UAVs, handle resource conflicts, and monitor progress.
Agronomist	The process of surveying crops, prescribing medication, and testing active ingredients.
Technician	UAV maintenance, battery management, handling technical alerts.
Administrator	Account management, security, and system monitoring.

The results obtained

- Create a list of 25 Functional Requirements.
- Identify the 7 Business Rules.
- Complete the Use Case and define the roles of each Actor.

4.2 Document & Domain Research

The team conducted document and domain research to understand the specifics of the agricultural UAV service industry before modeling the requirements.

References

- Software Requirement Specification (SRS) document.
- IEEE Std 830-1998 on software requirements specifications.
- Studies on the cost and workload assessment of agricultural drone spraying services.
- Studies on the economic impact of UAVs in pesticide application.
- Regulations for UAV operation and aviation safety standards.
- Business documentation on equipment maintenance management and service management.

Research activities

- Analyze the service process from the moment the customer submits a request until payment is completed.
- Learn about the operational constraints of UAVs in the field.
- Research on UAV equipment maintenance procedures and resource management for mission coordination.
- Compare the legal requirements related to the banned substance and the operating certificate.

The results obtained

- Xác định System Boundary (In Scope / Out of Scope).
- Establish business rules and assumptions.
- Develop NFR requirements for performance, security, readiness, and auditability.

4.3 Scenario Analysis

The team used Scenario Analysis to simulate real-world operating scenarios and determine the expected system response.

Target

- Check the completeness of the request.
- Identify the primary thread, alternative thread, and exceptions.
- Add special processing conditions.

Scenario 1: Spraying schedule successfully booked.

1. The farmer logs into the system.
2. Create a land plot profile.
3. Please submit your service request along with information about your plants, pests, and suggested pesticides (if any).
4. The manager reviews the request.
5. The agronomist assesses the requirements and creates a spray prescription.
6. Manager coordinates UAVs and drone operators.
7. Drone operator performs mission.
8. Recording system Operation Log and Spray Log.
9. Create invoices and complete orders.

Result:

Sinh FR-05, FR-09, FR-22, FR-23.

Scenario 2: Schedule conflict and device unavailable

1. The manager assigned the UAV to a new mission.
2. The UAV detection system is undergoing maintenance.
3. The system refused the assignment.
4. The dashboard displays a warning.

Result:

Born FR-06, FR-08, FR-14 and BR-03.

Scenario 3: Internet connection lost at the scene.

1. Operator downloads data offline.
2. Keep a field log.
3. The data is stored locally.
4. Synchronizes with the system when an internet connection is available.

Result:

Born FR-18 and NFR-08.

Scenario 4: Prescription containing a banned substance

1. The farmer submitted a request along with a suggested pesticide.
2. The agronomist assesses the request and creates a prescription.
3. Chemical catalog verification system.
4. The active ingredient, which is on the blacklist, has been detected.
5. Refuse to approve the prescription.

Result:

Born FR-11 and BR-02.

Through Scenario Analysis, the team verifies the consistency between the system's Functional Requirements, Business Rules, and Use Cases.

5. System boundary and assumptions

5.1 System Boundary

Agriculture UAV is a platform for managing and coordinating precision agricultural spraying drone services. The system focuses on supporting the management of the entire UAV service process, including the following functions:

In Scope

- User account management and role-based access control.
- Managing land plot records and user service requests.
- Managing and developing electronic prescriptions for pesticide spraying operations.
- Mission coordination, including personnel assignment and UAV resource management.
- Monitor task progress throughout the operational process.
- Recording operational logs includes: OperationLog and Spray Log
- Manage maintenance operations for UAV equipment, batteries, and charging stations.
- Managing invoices and compiling reports to support business operations.

Out of Scope

- Direct flight control or automated UAV flight control.
- AI systems for pest and disease identification or agricultural image analysis.
- Processing real-time telemetry data from UAVs.
- Managing UAV firmware or hardware systems.

5.2 Assumptions

- System users are assumed to have valid accounts and be assigned the correct permissions according to their roles within the system.
- The farmer provides complete and accurate information regarding the land plot and service requirements.
- Any pesticides proposed by the Farmer (if any) will be reviewed and approved by the Agronomist before being included in the spraying plan.
- Map data and weather information are provided from third-party services and are guaranteed to be reliable.

- Drone operators can record missions offline when there is no network connection and synchronize data when the connection is restored.
- The UAV equipment, batteries, and operating personnel are managed and ensured to be always operational and ready for use by the company.

6. Functional Requirements

FR ID	Requirement (Features)	Priority	Input (Input data)	Processing / Business Rule	Output (Output data)
FR-01	Admin manages accounts centrally.	Must	User profile (Name, Email, Role)	Check for required and duplicate Email/Username data on the system using the RBAC permission model.	A new account has been created, and a temporary password has been sent to your email.
FR-02	Monitoring operations and troubleshooting	Must	Log ID, Date Filters, Actor ID	The system automatically logs changes to structured data. Only administrators have access to check for errors.	The list of system impact history is displayed visually.
FR-03	Log in and authenticate security.	Must	Email, Password	Encrypting login information, authenticating sessions, and navigating users based on roles.	Successfully accessed the corresponding Dashboard interface.
FR-04	Farmer manages land records.	Must	Field Location, Crop Type	This allows farmers to use a mobile app to create farmland profiles by pinning locations on a map or using the device's current location. The	Farmers' land records have been successfully stored on the system.

				system stores location information, crop types, and farming history. Location and boundary information can be verified or updated through field surveys by drone operators or technical staff.	
FR-05	Farmers schedule pesticide spraying services.	Must	Selected Field, Desired Date, Crop Information, Pest Symptoms, Preferred Pesticide (Optional)	Verify the Farmer's valid land records and save information on crops, pests, and recommended pesticides (if any). The system will set the initial status of the service request to Pending Approval.	New service requests are initiated and sent directly to the Manager.
FR-06	Automatically check and detect calendar conflicts.	Must	Mission ID, Schedule Datetime	The system automatically scans the pilot's duty roster and the UAV's operational schedule to detect overlaps, providing visual alerts.	Conflict warnings are displayed if scheduling conflicts are detected; this prevents the assignment of duplicate resources.
FR-07	Integrate and display large-scale no-fly zone maps.	Must	Flight Location, Flight Coordinates	Configure and display large-scale no-fly zone data layers on the Manager's dispatch map to avoid being assigned to	No-Fly Zones are visually marked in red on the navigation map.

	(Ministry of Defence, 2026)			hazardous areas.	
FR-08	Display resource availability/busy status.	Must	Resource ID, Time Filter	Scan and display the schedule and Ready/Busy/Maintenance status of all pilots, drones, and batteries on the dashboard.	Visual scheduling charts assist managers in making quick coordination decisions.
FR-09	Real-time task progress monitoring	Must	Mission ID, Status updates	Consolidate and display the operational status of flight crews in the field according to their workflow on a centralized platform.	The dashboard screen provides continuous progress updates.
FR-10	Drug invoice number and field report	Must	Mission ID, Pesticide type, Dosage, Water volume	Agronomist reviews the crop information, pests and diseases, and pesticides recommended by the Farmer (if any). The system checks the suitability of the pesticide, the dosage, and related regulations before generating an electronic prescription.	Electronic prescriptions are approved and attached directly to the assignment file.
FR-11	Regulatory Filter	Must	Pesticide ID, Prescription Data	The system automatically compares the active ingredients in the prescription with the Ministry of Agriculture and Rural Development's list of banned/restricted chemicals.	Automatically block and alert if a prescription contains an active ingredient on the blacklist.

FR-12	Manage your Drone and Battery inventory.	Must	Drone Model, Serial Number, Specs	Add new devices to the system and track the accumulated flight hours of each device, as well as the number of charge/discharge cycles of each battery.	The catalog of UAV equipment and batteries is stored and managed centrally.
FR-13	Manage and retrieve repair history.	Must	Drone ID, Component replaced, Test result	Record repair costs and history. When the UAV is in Under Maintenance mode, the locking system prevents it from being assigned to a new flight schedule.	The UAV's movement status is correct (Ready or In Maintenance).
FR-14	Automatically issue scheduled maintenance alerts.	Must	Flight Hours threshold, Battery Cycle threshold	The system allows for configuring maintenance thresholds for each UAV model and battery type to ensure scalability and compliance with manufacturer standards.	Scheduled maintenance notifications are automatically sent to the technical department.
FR-15	ValidationAir pilot license (Government, 2025)	Must	Operator ID, License Expiry Date	The automated system checks the validity period of the drone operator's flight certificate (meeting the standards): + For aircraft weighing 0.25 kg or more: a Visual Flight Control Certificate (Class A) is required. + For aircraft weighing 2 kg or more, or flying beyond visual range: a Class B Instrument Flight Operator's License issued	Automatically lock assignment privileges and issue warnings if the pilot's flight certification is inappropriate.

				by the Ministry of Defense or an international certificate recognized by the Ministry of Defense is required.	
FR-16	Fill out the pre-flight safety checklist.	Must	Drone ID, Battery ID, Safety Checklist	Provide a digitized form for operators to check the static hardware safety checklist (propellers, frame, battery > 90%) on-site before takeoff.	The UAV was electronically approved and switched to the Ready state to perform its mission.
FR-17	Operating mobile applications offline	Must	Offline Cache Request, Mission ID	This allows the drone operator to pre-download prescription data and flight area maps to local storage for operation in areas without internet access.	Ready maps and mission information can be used offline on mobile devices.
FR-18	Record data in the Spray Log.	Must	Area covered, Payload sprayed, Wind speed	The system provides a digitized form for operators to manually fill in the actual spray parameters after landing. Once saved, the log data cannot be edited directly. Any changes must go through the update request and audit log process.	The spray log has been successfully initialized, ensuring data integrity.
FR-19	Record the UAV Operation Log.	Must	Flight duration, Battery usage data	The system automatically calculates the UAV's flight hours and updates them cumulatively in the equipment's record, while	Flight logOperationLog records are recorded and synchronized to

				also logging the data for use by environmental management agencies.	the database.
FR-20	Transmit emergency signals and update emergency status.	Must	Mission ID, Emergency Type, Description	This allows the operator to quickly input estimated coordinates and manually describe the emergency incident in the field (drone crash, remote control signal loss) to request a postponement/cancellation of the scheduled event.	Emergency (SOS) alerts with descriptions appear immediately on the Manager Dashboard.
FR-21	Track your order status visually.	Must	Farmer Account ID, Mission ID	It provides an intuitive progress interface that helps farmers track service request status, prescription validation processes, and task execution.	Order status is displayed transparently on Farmer's Web/Mobile interface.
FR-22	Electronic service invoice management	Must	Mission ID, Base rate, Actual area	Automatically calculates costs based on the actual sprayed area and Spray Log data to generate a service invoice (VAT, surcharges, distance if applicable).	Electronic invoices awaiting payment are issued and sent to the Farmer.
FR-23	Record manual payment confirmation	Should	Invoice ID, Payment Method	Record payment information (cash or bank transfer) from the Farmer to update the invoice status to Paid and close the task.	The invoice status changes to Paid, manually storing the financial transaction history.

FR-24	Record feedback on service quality.	Optional	Mission ID, Rating (1-5 stars), Feedback	Allows farmers to rate the quality of the spraying service and the attitude of the operating team after the order is completed.	The evaluation score is recorded and saved in the Drone Operator's profile.
FR-25	Compile revenue reports and export local flight schedule data.	Should	Date filters, Region filters	Compile data from paid invoices to create revenue statistics charts; support exporting flight schedule data forms for declaration to local authorities.	The dashboard chart is visually displayed; the report data file has been successfully exported (Excel/PDF).

7. Non-Functional Requirements

NFR ID	Category	Requirement	Measurement
NFR-01	Performance	Data loading speed on the management dashboard	<= 2 seconds for 1,000 historical data records(Google, 2023).
NFR-02	Performance	Dashboard chart synchronization processing speed	<= 3 seconds when the Manager performs filtering and aggregation of annual report data. (Replacing the IoT trap)(Amazon Web Services, 2024).
NFR-03	Security	Security features require authentication.	100% of internal business functions require login, except for the public introductory page.(NIST, 2020)
NFR-04	Security	Encrypting stored data and securing information.	100% of passwords must be stored as hashes using industry-standard algorithms.(Provos & Mazières, 1999)

NFR-05	Usability	Mobile app interface visibility in sunlight.	A screen contrast ratio of $\geq 4.5:1$ allows pilots to easily operate the controls even when operating the controls in rice paddies.(W3C, 2023)
NFR-06	Usability	Increase the accessibility of the system for new operators.	The system must provide a basic operation tutorial video, with a maximum length of 10 minutes, ensuring that new operators can proficiently download and view offline maps after watching the video..(Nielsen, 1994)
NFR-07	Availability	Cloud platform readiness	Achieve an uptime rate of $\geq 99.9\%$, operating continuously 24/7 (except for scheduled system maintenance as announced in advance).(Amazon Web Services, 2024).
NFR-08	Robustness	Ability to retain data on mobile apps when 3G/4G signal is lost.	The goal is to ensure that 99.9% of manually entered log data is securely stored temporarily in the device's local storage and automatically uploaded to the server as soon as a stable connection is established.(Ningrum et al., 2026).
NFR-09	Auditability	Log system transactions for critical business operations.	The system automatically and permanently logs user IDs, timestamps, IP addresses, and action details for service request approvals, spray prescription creation or updates, task approvals, and other critical administrative operations.(NIST, 2020)
NFR-10	Performance	Delay in updating task status on the real-time monitoring dashboard.	Mission status update delay ≤ 5 seconds from the time the system receives the update data.(Amazon Web Services, 2024)

8. Business Rules

Rule ID	Business Rule
BR-01	Farmers must schedule the service at least 48 hours before their scheduled flight.
BR-02	The system switches to "Delay" mode if the wind speed is > 5 m/s or if it is raining.
BR-03	UAVs are only assigned schedules when they are in "Ready" mode and not undergoing maintenance.
BR-04	Flight schedules are only assigned after the mission has an electronic prescription.
BR-05	The system will alert you to additional charges or request confirmation if you exceed the service radius.
BR-06	Canceling your flight less than 12 hours before departure will automatically incur a 10% penalty on the basic fare.
BR-07	If the actual flight time deviates by more than 15% from the planned time, the system automatically flags the mission for the Manager to review.
BR-08	All pesticides proposed by the Farmer must be reviewed and approved by the Agronomist before being included in the spraying plan.
BR-09	The system automatically switches the mission to "Suspended" and locks the UAV's status to "Under Maintenance" as soon as it receives an emergency (SOS) signal while in flight.
BR-10	When a mission is interrupted, the system automatically fixes the sprayed volume and generates a replacement flight schedule for the remaining area without charging any additional basic fees.
BR-11	The system blocks invoicing and prohibits changing the duty assignment to "Completed" for flights with incidents until the Technician or Manager confirms that the issue has been resolved.
BR-12	The system blocks service requests if the land parcel coordinates are located within a No-Flight Zone (NFZ).
BR-13	The takeoff system is blocked (status "In Progress") if any item in the pre-flight safety checklist is marked "Fail".
BR-14	The system will only issue invoices once the offline spray log has been 100% synchronized to the server.

BR-15	The system automatically blocks prescription approvals and issues violation warnings if any active ingredient in the prescription is on the Ministry of Agriculture and Rural Development's list of banned/restricted chemicals.
BR-16	The system blocks the assignment of missions to drone operators if the flight control license is not in the correct category according to the UAV's takeoff weight (Category A: $\geq 0.25\text{kg}$ for visual flight; Category B: $\geq 2\text{kg}$ or for out-of-visual-range flight) or if the license has expired.
BR-17	The email/username of the account must be unique across the entire system; new accounts can only be created with a valid role according to the RBAC model configured by the Administrator.
BR-18	The land boundaries declared by the Farmer on the map are in "Unverified" status; they can only be used for official flight planning after confirmation by the Drone Operator or Technician through a field survey.
BR-19	Once recorded, Spray Log data is immutable; any modification requests must go through a separate approval process and be logged to System_Audit_Log. Direct modification/deletion of the original log is not permitted.
BR-20	Only administrators have access to System_Audit_Log to view the history of system operations and errors.
BR-21	The system automatically detects and blocks scheduling conflicts if the drone operator or UAV has already been assigned to another task with overlapping time slots.
BR-22	When the UAV's accumulated flight hours or battery charge/discharge cycles exceed the configured threshold for each device model, the system automatically switches to "Maintenance Required" status and blocks it from being assigned to a new flight schedule until a technician confirms that maintenance is complete.
BR-23	Offline mission data (prescriptions, flight area maps) are temporarily stored on the mobile device's local memory and must be automatically synchronized to the server as soon as a stable network connection is restored.
BR-24	Electronic prescriptions are only approved after Agronomist checks and confirms the suitability between the recommended pesticide type and dosage and the actual pest/crop condition.
BR-25	Changing your personal account password requires you to enter your current password; the system will reject the update if verification fails.

9. Use Case List

UC ID	Use Case Name	Primary Actor	Priority
UC-01	Centralized account management	Administrator	Must
UC-02	Monitoring operations and troubleshooting	Administrator	Must
UC-03	Log in and authenticate security.	All the Actors	Must
UC-04	Land map and record management	Farmer	Must
UC-05	Schedule agricultural spraying services.	Farmer	Must
UC-06	Coordinate flight schedules and monitor resources.	Manager	Must
UC-07	Assess medication requests and prescribe medications.	Agronomist	Must
UC-08	Emergency response	Drone Operator	Must
UC-09	Manage your Drone and Battery inventory.	Technician	Must
UC-10	Automated maintenance management and alerts	Technician	Must
UC-11	Verify the pilot's drone flight license.	Manager	Must
UC-12	Fill out the pre-flight safety checklist.	Drone Operator	Must
UC-13	Operating mobile applications offline	Drone Operator	Must

UC-14	Record flight/spray data and logs.	Drone Operator	Must
UC-15	Track your order status visually.	Farmer	Must
UC-16	Invoice management and payment control	Farmer / Manager	Must
UC-17	Record feedback on service quality.	Farmer	Optional
UC-18	Compile sales reports and export data.	Manager	Should

10. Use Case Diagram

The overall Use Case diagram models the system boundaries, actors, and how they interact with the functional subsystems. **Drone Coordination & Monitoring System.**

101. List of Actors (User Classes / Actors)

The system comprises 6 layers of actors (4 core business actors, 1 technical support actor, and 1 administrative actor):

- **Farmer (FM):**The customer/farmer submits a service request.
- **Drone Operator (OP):**Pilots directly operate the aircraft in the field.
- **Technician (TC):**Maintenance technicians perform safety inspections and maintenance on hardware equipment.
- **Manager (MG):**Managing and coordinating the entire workflow.
- **Agronomist (AG):**Agricultural experts are responsible for assessing spraying requirements and prescribing pesticides.
- **Administrator (AD):**System administrator, software infrastructure manager.

10.2. System Boundary and Interactions

The whole **17 Use Cases** It is enclosed within the System Boundary and divided into four main subsystems as follows:

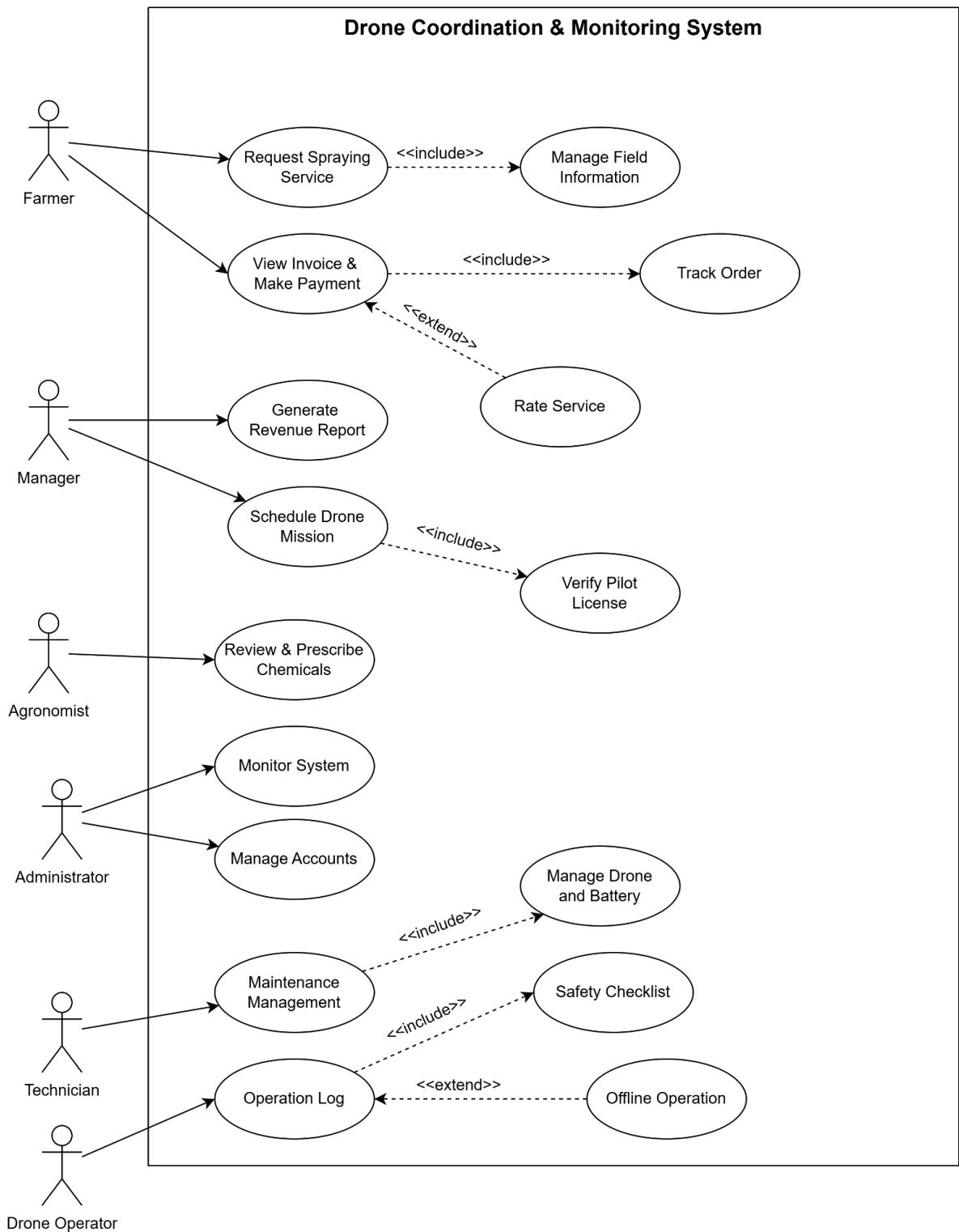
UC ID	Use Case Name	Primary Actor	System relationships
UC-01	Centralized account management	Administrator	Operates independently

UC-02	Monitoring operations and troubleshooting	Administrator	Operates independently
UC-03	Log in and authenticate security.	All Actors	Preconditions for the entire system
UC-04	Land map and record management	Farmer	<<include>> by UC-05
UC-05	Schedule a pest control service.	Farmer	<<include>> UC-04
UC-06	Coordinate flight schedules and monitor resources.	Manager	<<include>> UC-11; <<extend>> by UC-07
UC-07	Assess medication requests and prescribe medications.	Agronomist	<<extend>> UC-05; precursor to UC-06
UC-08	Emergency response	Drone Operator	
UC-09	Manage your Drone and Battery inventory.	Technician	<<include>> by UC-10
UC-10	Automated maintenance management and alerts	Technician	<<include>> UC-09
UC-11	Verify the pilot's drone flight license.	Manager	<<include>> by UC-06
UC-12	Fill out the pre-flight safety checklist.	Drone Operator	<<include>> by UC-14
UC-13	Operating mobile applications offline	Drone Operator	<<include>> by UC-14
UC-14	Record flight/spray data and logs.	Drone Operator	<<include>> UC-12, UC-13
UC-15	Track your order status visually.	Farmer	<<include>> by UC-16
UC-16	Invoice management and payment control	Farmer / Manager	<<include>> UC-15; <<extend>> by UC-17
UC-17	Record feedback on service quality.	Farmer	<<extend>> UC-16
UC-18	Compile sales reports and export data.	Manager	Operates independently

Note

- **<<include>>**: A function that must be performed.
- **<<extend>>**: An extended function or one that occurs after a business process step.
- **UC-03 Login and security authentication** It is considered a prerequisite for the entire system and does not require the use of relationships. <<include>> Include all use cases to avoid making the diagram too complicated.

10.3. Logical Structure Diagram



10.4. Important Constraints When Converting to Figure Drawings

1. Agent Arrangement

- All Actors are placed outside the system boundaries.
- The Actors are arranged in order of their professional roles from top to bottom:
 - Farmer
 - Agronomist
 - Manager
 - Drone Operator
 - Technician
 - Administrator
- This order reflects the system's business process flow.

2. Association Relationship

- Use a solid line between the Actor and the Use Case.
- Each Actor only connects to the functions they directly use.
- Do not draw direct relationships between the Actors.

3. The <<include>> relationship

Relationship **<<include>>** It is used when a Use Case always requires performing another function as a mandatory step.

For example:

- Flight schedule coordination **<<include>>** Verify the pilot's drone flight license.
- Record flight logs. **<<include>>** Safety checklist.
- Payment management **<<include>>** Track your order.

4. The <<extend>> relationship

Relationship **<<extend>>** It is used when the function only arises in certain cases or at the next stage of the process.

For example:

- Requirements assessment **<<extend>>** Schedule an appointment.
- Service evaluation **<<extend>>** Pay.

5. Log in and authenticate.

Function **Login and secure authentication (UC-03)** It is considered a prerequisite for the system.

Because all users must log in before using business functions, UC-03 cannot use relationships.<<include>>with all the Use Cases to avoid making the diagram complicated and difficult to follow.

This condition is described in the Use Case Specification:

Pre-condition: The user has been successfully authenticated before performing the function.

6. System boundaries

All use cases are contained within the System Boundary:**Drone Coordination & Monitoring System**

Use Case diagrams focus on illustrating the relationships between actors and business processes, without dividing them into functional modules, to ensure visual clarity and compliance with the UML Use Case Diagram standard.

11. Detailed Use Case Description

11.1. UC-05 Detailed Specification: Creating a Spraying Service Request

Item	Description
Use Case ID	UC-05
Use Case Name	Schedule a pest control service.
Primary Actor	Farmer
Goal	The farmer submits a request for pesticide spraying services, including information about the crops, pests, and treatment needs for a specific area of land.

Preconditions	• The farmer has successfully logged into the system (UC-03). • The farmer has at least one previously stored farm land record (UC-04).
Trigger	The farmer selects the "Create spray request" function on the interface.
Main Flow	1. The system displays the request creation form and a list of the Farmer's existing farmland records. 2. The farmer selects the area of land that needs to be sprayed with pesticides. 3. The farmer enters the desired date and time for spraying, describes the pest and disease situation, and suggests the desired pesticide. 4. The farmer clicks the "Submit Request" button. 5. System for checking scheduling time constraints (BR-01). 6. The system saves the request to the database with an initial status of Pending Approval. 7. The system displays a message confirming the request was successfully created and sends a notification to the Manager.
Alternative Flow	5a. The scheduled flight time violates the binding rules: 1. The Farmer's time detection system selects a time less than 48 hours from the current time. 2. The system refused to save and displayed the error message: "Please

	schedule at least 48 hours in advance." 3. Return to step 3 of the Main Flow.
Postconditions	A new service request has been successfully initiated on the system with the status Pending Review.
Business Rules	BR-01 (Minimum booking time: 48 hours).

11.2. UC-06 Detailed Specification: Flight Schedule Coordination and Resource Monitoring

Item	Description
Use Case ID	UC-06
Use Case Name	Coordinate flight schedules and monitor resources.
Primary Actor	Manager
Goal	Finalize flight schedules by assigning the appropriate Drone Operator and UAV equipment to the approved mission, avoiding scheduling conflicts and no-fly zones.
Preconditions	• Manager has successfully logged in (UC-03). • The mission has been approved and an updated prescription has been issued by Agronomist (UC-07, BR-04).

Trigger	The manager selects a task with the Prescribed status and clicks "Coordinate Flight Schedule".
Main Flow	1. The system displays mission information, no-fly zone data class (FR-07), and a list of available Drone Operators and UAVs during that time slot. 2. The manager selects a Drone Operator and a UAV device from the list. 3. The manager clicks "Confirm Coordination". 4. System for monitoring UAV operating status (BR-03) and weather forecast data for the flight area (BR-02). 5. Operator and UAV scheduling check system. 6. The system updates the mission status to Assigned and finalizes the flight schedule. 7. The system automatically sends work schedule notifications to the assigned Drone Operator's mobile application.
Alternative Flow	4a. The selected UAV is unqualified or undergoing maintenance: 1. The system detects that the selected UAV is in a state other than "Ready" or is on a static maintenance schedule. 2. The system displays an error warning and does not allow scheduling. 3. Return to step 2 of the Main Flow. 4b. Severe weather warning activation system: 1. The weather monitoring system at the plot location detects wind speed > 5 m/s or rain. 2. The system displays a bad weather warning icon and blocks the booking confirmation. 3. The manager will discuss with the farmer to reschedule the flight to a different time. 5a. Detecting scheduling conflicts: 1. The system detects that the operator or UAV has been assigned to a

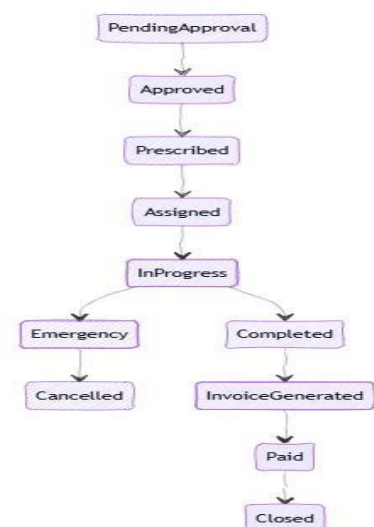
	different task within the same time slot. 2. The system displays an error message: "Schedule Conflict". 3. Return to step 2 of the Main Flow.
Postconditions	The task has been changed to the Assigned status. The work schedule has been synchronized with the Drone Operator.
Business Rules	• BR-02 (Warning and schedule blocking during bad weather). • BR-03 (UAV must be in "Ready" mode and not undergoing maintenance). • BR-04 (A prescription is required before scheduling an appointment).

11.3. State Transition Diagrams

11.3.1. State Diagram – Mission Lifecycle (FR-09, FR-21, FR-22, FR-24)

describe **Flight mission lifecycle**
link:

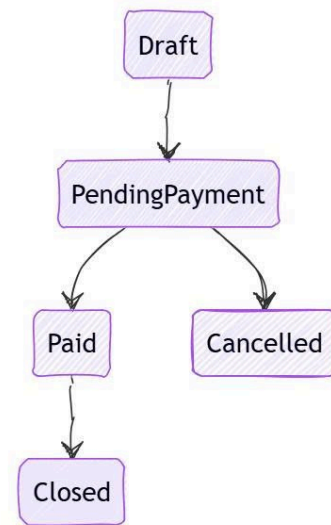
- FR-09 Mission Progress Monitoring
- FR-20 SOS
- FR-21 Order Status Tracker
- FR-23 Payment → Mission Closed



11.3.2. State Diagram – Invoice Payment (FR-23, FR-24, FR-25)

describe **Invoice lifecycle**
link:

- FR-22 Invoice Management
- FR-23 Payment Confirmation
- FR-25 Sales Report



12. Data Requirements

This section defines the logical data structures (Data Entities) that the system must store to support the features in D2 and prepare for the ERD model in D3.

12.1. Land Records Entity (Fields)

- Purpose: To store geographical information of the Farmer's land plot (FR-04).
- Core attributes:
 - field_id (Primary Key - String): Land parcel identifier.
 - farmer_id (Foreign Key -String): Link to Farmer account.
 - field_name (String): The name of the land plot (e.g., Thuong Field 01).
 - crop_type(String): Crop type (Rice, Fruit tree, etc.).
 - boundary_coordinates (Geometry/Text): The sequence of GPS coordinates that form a closed polygon drawn from a satellite map.
 - calculated_area(Float): The area of the land plot is automatically calculated (unit: m²).²).

12.2. Flight Mission Entity

- Purpose: To manage the lifecycle of a spraying appointment from creation to completion (FR-05, FR-06).
- Core attributes:
 - mission_id(Primary Key - String): Mission Code.
 - field_id (Foreign Key - String): The land plot being implemented.
 - manager_id (Foreign Key -String): Schedule Coordinator.
 - operator_id(Foreign lock -String): Assigned pilot.
 - drone_id (Foreign Key - String): Assigned UAV device.

- `scheduled_start_time` (Datetime): The scheduled takeoff time.
- `status` (Enum): Task status (*Pending Approval, Approved, Prescribed, Assigned, In Progress, Completed, Closed*).

12.3. Electronic Prescription Entity (Prescriptions)

- **Purpose:** To digitize the technical information on pesticide application prescribed by Agronomists (FR-10).
- **Core attributes:**
 - `prescription_id` (Primary Key - String): Prescription ID.
 - `mission_id` (Foreign Key - String): Which task does it belong to?
 - `medicine_name` (String): Name of the pesticide.
 - `Dosage` (Float): Specific dosage.
 - `dilution_ratio` (String): The dilution ratio with water.

12.4. The User Entity

- **Purpose:** Manage account information and assign permissions to all users in the system (FR-01, FR-03).
- **Core attributes:**
 - `user_id` (Primary Key - String): User identifier.
 - `full_name` (String): User's full name.
 - `email` (String): The email address used to log into the system.
 - `password_hash` (String): The encrypted password.
 - `role` (Enum): User role (Administrator, Manager, Farmer, Agronomist, Drone Operator, Technician).
 - `phone_number` (String): Contact phone number.
 - `account_status` (Enum): Account status (Active, Inactive, Suspended).
 - `created_at` (Datetime): The time the account was created.

12.5. UAV Device (Drone) Entity

- **Purpose:** Manage the inventory of UAV equipment used for flight mission coordination and operation (FR-13, FR-14, FR-15).
- **Core attributes:**
 - `drone_id` (Primary Key -String): UAV identifier.
 - `serial_number` (String): The serial number of the device.
 - `model_name` (String): Tên model UAV.
 - `drone_status` (Enum): Device status (Ready, Busy, Under Maintenance).
 - `accumulated_flight_hours` (Float): Total accumulated flight hours.
 - `purchase_date` (Date): The date the device was put into use.
 - `last_maintenance_date` (Date): The date of the last maintenance.

12.6. EntitiesgUpon receiving the UAV Operation Log.

- **Purpose:** Storing the operational history and flight times of the UAV for auditing and equipment maintenance (FR-20).
- **Core attributes:**
 - log_id (Primary Key - String): Flight log ID.
 - mission_id (Foreign Key - String): Related task.
 - drone_id (Foreign Key - String): The UAV performing the flight.
 - operator_id (Foreign lock -String): Pilot in control.
 - flight_duration (Float): Actual flight duration (minutes).
 - battery_usage (Float): Battery consumption during flight.
 - flight_date (Datetime): The flight time.
 - Remarks (Text): Additional notes.

12.7. Service Invoice Entity

- **Purpose:** Manage payment information and revenue from completed tasks (FR-23, FR-24, FR-25).
- **Core attributes:**
 - invoice_id (Primary Key - String): Invoice ID.
 - mission_id (Foreign Key - String): This task is subject to a fee.
 - farmer_id (Foreign Key - String): Customer payment.
 - total_amount (Decimal): The total value of the invoice.
 - payment_method (Enum): Payment method (Cash, Bank Transfer).
 - invoice_status (Enum): Invoice status (Pending, Paid, Cancelled).
 - issue_date (Datetime): The invoice issuance date.
 - payment_date (Datetime): Payment confirmation date.

12.8. Maintenance Records Entity

- **Purpose:** Track the maintenance, repair, and inspection history of UAV equipment (FR-14, FR-15).
- **Core attributes:**
 - maintenance_id (Primary Key - String): Maintenance ticket ID.
 - drone_id (Foreign Key - String): The UAV device is under maintenance.
 - technician_id (Foreign Key - String): Technician performing the work.
 - maintenance_type (Enum): Maintenance type (Preventive, Corrective).
 - maintenance_date (Datetime): The date the maintenance was performed.
 - maintenance_cost (Decimal): Maintenance cost.
 - maintenance_result (Text): Test and repair results.
 - next_maintenance_date (Date): The next scheduled maintenance date.

12.9. Operator Licenses Entity

- **Purpose:**Manages the validity of UAV operator certificates to support the flight authentication and assignment process (FR-16).
- **Core attributes:**
 - license_id (Primary Key -String): Certificate identifier.
 - operator_id(Foreign lock -String): Link to Drone Operator account.
 - license_number (String): UAV operator license number.
 - issued_date (Date): Date the certificate was issued.
 - expiry_date (Date): Certificate expiration date.
 - issuing_authority(String): The certificate issuing authority.
 - license_status (Enum): Certificate status (Valid, Expired, Suspended).

12.10. Spray Logs Entity

- **Purpose:**Record actual spraying data after task completion for operational monitoring, history retrieval, and service cost calculation (FR-19, FR-23).
- **Core attributes:**
 - spray_log_id (Primary Key - String): Spray log ID.
 - mission_id(Foreign Key - String): Related task.
 - operator_id(Foreign lock -String): The pilot is performing the mission.
 - area_covered (Float): Actual sprayed area.
 - payload_sprayed (Float): The actual amount of pesticide sprayed.
 - wind_speed(Float): The wind speed recorded during the spraying process.
 - spray_datetime(Datetime): The time the spraying was carried out.
 - Remarks (Text): Additional notes from the Drone Operator.

12.11. Emergency Reports Entity

- **Purpose:**Storing emergency incidents reported by the Drone Operator during mission execution to support situation handling, operational monitoring, and incident history tracking (FR-21).
- **Core attributes:**
 - emergency_id(Primary Key -String): Emergency report identifier.
 - mission_id(Foreign Key - String): Task related to the incident.
 - operator_id(Foreign Key - String): Drone Operator sends a report.
 - emergency_type (String): Loại sự cố khẩn cấp (Drone Crash, Signal Loss, Battery Failure, Weather Emergency, Other).
 - Description (Text): Detailed description of the problem.
 - Latitude (Float): The latitude of the location where the incident occurred.
 - Longitude (Float): Longitude of the incident location.

- reported_at(Datetime): Time for submitting urgent reports.
- Status (Enum): Incident handling status (Reported, Investigating, Resolved).

12.12. Service Evaluation Entity (Reviews)

- **Purpose:** Storing Farmer reviews and feedback after service completion helps measure operational quality and improve customer experience (FR-25).
- **Core attributes:**
 - review_id (Primary Key - String): Review identifier.
 - mission_id (Foreign Key - String): The task being evaluated.
 - farmer_id (Foreign Key -String): The person conducting the evaluation.
 - Rating (Integer): Service quality rating (1–5 stars).
 - Feedback (Text): Customer comments and suggestions.
 - created_at (Datetime): Time the review was submitted.

12.13. UAV Battery Entity

- **Purpose:** Managing UAV battery information, monitoring charge/discharge cycles and usage status to support the operation and maintenance of the aircraft (FR-12, FR-14, FR-16).
- **Core attributes:**
 - battery_id (Primary Key -String): PIN identifier.
 - drone_id (Foreign Key - String): The UAV currently in use or assigned to a battery.
 - serial_number(String): The serial number of the battery.
 - cycle_count (Integer): Total number of charge/discharge cycles used.
 - battery_status (Enum): Trạng thái pin (Ready, Charging, In Use, Maintenance, Retired).
 - last_charged_at (Datetime): The time of the last full charge.

12.14. Service Request Entities

- **Purpose:** Record the Farmer's spraying needs (UC-05), which serves as input for the Manager to create corresponding missions. This acts as a bridge between actual needs and the operational plan.
- **Core attributes:**
 - request_id (Primary Key - String): Service request ID.
 - farmer_id (Foreign Key -String): The person making the request.
 - field_id (Foreign Key - String): The plot of land to be sprayed.

- `desired_date` (Datetime): The desired time for the service to be performed.
- `symptoms` (Text): Describe the condition of the plant or the reason for the request (e.g., leaf-rolling caterpillars, brown planthoppers...).
- `proposed_pesticide`(String): The type of pesticide that Farmer recommends (if any).
- `proposed_dosage`(String): Desired dosage or treatment method for the Farmer (if any).
- `status` (Enum): Request status (Pending, Under Review, Approved, Rejected, Converted_To_Mission).
- `created_at`(Datetime): The time the request was submitted.
- `Note` (Text): Additional notes from the Farmer.

12.15. System Audit Logs Entity

Purpose: Record all data modification and system access activities for monitoring, error tracing, and accountability purposes (FR-02).

Core attributes:

- `audit_id` (Primary Key - String): Log record ID.
- `user_id` (Foreign Key - String, can be null): The person performing the action; leave blank if it is an automated system action.
- `user_role`(String): The user's role at the time of execution (snapshot, does not update backward if the role changes later).
- `action_type` (Enum): Action type (Create, Update, Delete, Login, Logout, Export).
- `entity_name`(String): The name of the affected table/object.
- `entity_id`(String): The ID of the affected record.
- `old_value` / `new_value` (JSON): The state of the data before and after the change.
- `action_status`(Enum): Action result (Success, Failed).
- `performed_at` (Datetime): The time the log was recorded.

12.16. No-Fly_Zones Entity

Purpose: Stores the boundaries of no-fly/restricted airspace areas so the system can check for overlaps when the Farmer creates a service request or the Manager coordinates the mission (FR-07).

Core attributes:

- `nfz_id` (Primary Key - String): No-fly zone code.

- zone_name (String): The name of the zone.
- zone_type (Enum): Loại vùng (Restricted, Prohibited, Temporary).
- polygon_coordinates (Geometry/Text): Area polygon coordinates.
- authority_source(String): The authority that issued the regulation.
- effective_from / effective_to (Datetime): Validity period (leave effective_to blank if permanent).

12.17. Entity: List of Banned Chemicals

Purpose: To serve as an automated reference point when Agronomist prescribes medication, ensuring that no banned active ingredient (FR-11) is used.

Core attributes:

- chemical_id (Primary Key - String): Active ingredient code.
- chemical_name (String, Unique): The name of the banned substance.
- active_ingredient(String): Active ingredient.
- regulatory_reference(String): Decision/circular number.
- banned_since(Date): The date the ban takes effect.

12.18. Preflight Safety Checklist Entity (Preflight_Checklists)

Purpose: Record the results of the mandatory safety check for the Drone Operator before each flight (FR-16).

Core attributes:

- checklist_id (Primary Key - String): Checklist ID.
- mission_id (Foreign Key - String, Unique): Related mission (1-1).
- operator_id (Foreign Key - String): Pilot performs inspection.
- battery_check, weather_check, payload_check, gps_signal_check (Boolean): Items to check.
- overall_status (Enum): Overall result (Passed, Failed).
- checked_at(Datetime): The time of checking.

12.19. Entity Relationships (Logical Data Model)

The system has the following main relationships:

- User (1) → (N) Fields
- Field (1) → (N) Missions
- Field (1) → (N) Service Requests
- Mission (1) → (1) Prescription

- Mission (1) → (1) Invoice
- Mission (1) → (N) Operation Log
- Mission (1) → (N) Spray Logs
- Mission (1) → (N) Emergency Reports
- Mission (1) → (N) Reviews
- Drone (1) → (N) Missions
- Drone (1) → (N) Maintenance Records
- Drone (1) → (N) Batteries
- Operator (1) → (N) Missions
- Operator (1) → (N) Licenses
- Operator (1) → (N) Operation Log
- Operator (1) → (N) Spray Logs
- Operator (1) → (N) Emergency Reports
- Farmer (1) → (N) Fields
- Farmer (1) → (N) Invoices
- Farmer (1) → (N) Reviews
- Farmer (1) → (N) Service Requests
- Service Request (1) → (0..1) Mission (Each request can be transformed into a mission, or be rejected).
- User (Any Role) (0) → (N) System_Audit_Logs
- Field (N) ↔ (N) No_Fly_Zones (check for overlap, do not save hard FK)
- Mission (1) → (1) Preflight_Checklists
- Operator (1) → (N) Preflight_Checklists
- Prescription (N) → (1) Banned_Chemicals (reference comparison, validation at the business layer)

13. External Interfaces

The system requires integration with the following three external application programming interfaces (APIs) for operation:

13.1. Weather Forecasting API Interface

- Integration partner: OpenWeatherMap API.
- Function: The system automatically sends the GPS coordinates (latitude, longitude) of the land parcel to the API 30 minutes before the flight.
- Data returned: Wind speed (wind_speed), rainfall. If wind speed exceeds 5 m/s or it rains, the system activates the rule. **BR-02**It automatically locks the task status and sends alerts to the Manager's control panel. This mechanism directly supports...**UC-06 (Flight Schedule Coordinator)**and related to the testing logic of**FR-06** and **FR-09**.

13.2. Geospatial & No-Fly Zone Interface

- Integration partners: Mapbox SDK / Google Maps API.

- Function: Uploads high-resolution satellite map layers to the Web/Mobile interface for the Farmer to draw land boundaries (FR-04). Simultaneously, integrates No-Fly Zone (NFZ) data from the aviation authority to display no-fly zones (airports, military bases) on the Manager's control screen (FR-07).

13.3. Manual Payment Recording Interface

- Integration partners: Not applicable in the current scope (the system does not integrate electronic payment gateways and does not process online transactions).
- Function:
The system supports manual recording of payments from Farmers via cash or off-system bank transfers. After verifying the actual transaction, the Manager confirms the payment on the management interface to update the invoice status to Paid according to FR-24. Payment data is stored for reconciliation, transaction tracing, and financial reporting purposes.

14. Requirement Traceability Matrix

FR ID	Use Case ID	Use Case Name / System Business Process	UI Screen ID & Screen Name	Related Model / Technical Proof
FR-01	UC-01	Centralized account management	UI-01 Account Management	RBAC Model, ERD: Users Table
FR-02	UC-02	Monitoring operations and troubleshooting	UI-02 System Log	Audit Log Design, Class Diagram
FR-03	UC-03	Log in and authenticate security.	UI-03 Login Screen	UML: Sequence Diagram
FR-04	UC-04	Land map and record management	UI-04 Land Map and Record Management	ERD: Fields Table

FR-05	UC-05	Schedule agricultural spraying services.	UI-05 Schedule Service	UML: Activity Diagram
FR-06	UC-06	Automatically check and detect calendar conflicts.	UI-06 Task Coordination	Use Case Diagram, Activity Diagram
FR-07	UC-06	Integration and display of wide-area no-fly zones (NFZ) maps (NFZ Basic Integration)	UI-06 Task Coordination	NFZ Map Data Structure, Layer Design
FR-08	UC-06	Display resource availability/busy status.	UI-07 Dashboard Monitoring	Dashboard UI, Class Diagram
FR-09	UC-06	Real-time task progress monitoring	UI-07 Dashboard Monitoring	Dashboard UI, Activity Diagram
FR-10	UC-07	Assess the request and create the electronic prescription.	UI-08 Electronic Prescription Device	Use Case Diagram, Class Diagram
FR-11	UC-07	Regulatory Filter	UI-09 Prohibited Substance Test	Business Rules, Chemical Blacklist
FR-12	UC-09	Manage your Drone and Battery inventory.	UI-10 Device Catalog Manager	ERD: Drones Table
FR-13	UC-10	Manage and retrieve repair history.	UI-11 Maintenance Management	Class Diagram, ERD
FR-14	UC-10	Automatically issue scheduled maintenance alerts.	UI-11 Maintenance Management	Business Rules, Activity Diagram

FR-15	UC-11	Verify the pilot's drone flight license.	UI-12 Drone Flight Permit Verification	Sequence Diagram, ERD: Operator_Licenses Table, Validation Rules
FR-16	UC-12	Fill out the pre-flight safety checklist.	UI-13 Pre-flight Safety Checklist	ERD: Drone_Checklists
FR-17	UC-13	Operating mobile applications offline	UI-14 Offline Mobile Application	UML: Component Diagram
FR-18	UC-14	Record data in the Spray Log.	UI-15 Flight Log and Spray	Use Case Specification, ERD
FR-19	UC-14	Record the UAV Operation Log.	UI-15 Flight and Spray Log	Audit Design, Database Design
FR-20	UC-08	Transmit emergency signals and update emergency status.	UI-16 Emergency Alert	UML: Sequence Diagram
FR-21	UC-15	Track your order status visually.	UI-17 Order Tracking	Use Case Diagram, Sequence Diagram
FR-22	UC-16	Electronic service invoice management	UI-18 Electronic Invoice	UI Prototype, ERD: Invoice, State Diagram: Invoice_Status
FR-23	UC-16	Record manual payment confirmation	UI-19 Payment Confirmation	Workflow Diagram, State Diagram: Payment_Status
FR-24	UC-17	Record feedback on service quality.	UI-20 Service Review	ERD: Reviews Table

FR-25	UC-18	Generate revenue reports and export local flight schedule data (Local Flight Declaration Tool)	UI-21 Dashboard Report	DFD: Analytics Flow, Dashboard UI, Revenue_Report Entity
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15. SRS Quality Checklist

The software requirements specification quality checklist below is based on the core criteria of the IEEE 830-1998 standard to assess the completeness of the D2 document before delivery to the system design phase.

Criteria on code	Quality Attributes	Status	Evidence in the system documentation.
QC-01	Correctness	Expected Outcome	100% of the functional requirements (FR) are directly broken down from the Farmer's field needs and the Manager's management process.
QC-02	Consistency	Expected Outcome	The identification system is standardized and used consistently throughout the document. Functional Requirements are numbered sequentially from FR-01 to FR-25, while Use Cases retain the same identifiers UC-01 to UC-18 to ensure the stability of references and traceability throughout the project lifecycle. There is no overlap or discrepancy in IDs between chapters.
QC-03	Completeness	Expected Outcome	The documentation covers all core modules committed in D1, including specific boundary constraints: Prohibited substance filters (FR-11), Pilot certificate verification (FR-16), and Offline Operating Mode (FR-18).

QC-04	Verifiability	Expected Outcome	Functional requirements (FR) are described with clear input data, processing rules, and output data. Key Use Cases (UC-05, UC-06) are specified in detail with Main Flow, Alternative Flow, Preconditions, and Postconditions, forming the basis for building Test Cases in subsequent testing phases.
QC-05	Unambiguous	Expected Outcome	Business rules are quantified using clear technical indicators rather than qualitative descriptions (e.g., a 48-hour advance scheduling requirement).BR-01The wind speed warning threshold is > 5 m/s atBR-02).
QC-06	Traceability	Expected Outcome	The Requirement Traceability Matrix (RTM) in Section 14 establishes an explicit two-way link between: Functional requirements↔Use Case Code↔ User interface (UI)↔Related technical model.
QC-07	Prioritization	Expected Outcome	All system requirements are clearly assigned weights according to the MoSCoW model. <i>Must, Should, Optional</i>) to optimize deployment resources in subsequent (Sprint) segments.

16. Conclusion and Next Steps

16.1. Conclusion

The Software Requirements Specification (SRS - Deliverable 2) document has completed the standardization of the entire project logic structure.**Drone-based agricultural spraying coordination and monitoring system.**Through the establishment of the Request Tracing Matrix (RTM), the project team hasContributing to ensuring consistency and minimizing discrepancies.The system has a functional interface between the problem definition phase (D1) and the detailed specification

phase (D2). The system already has a complete set of logical documentation ready to move to the architectural design and visual interface design phase.

16.2. Next Steps & Action Plan

To prepare for the Deliverable 3 (D3) document, the project team broke down the linear deployment roadmap into the following four core tasks, which will be directly applied to the members in the next segment:

1. Draw a Use Case Diagram (Overall & Subsystem)
2. Phân rã DFD (Context Level to Level 1 to Level 2)
3. Design the ERD Database (Define PK/FK, attributes)
4. Figma UI/UX Design (21-screen system from UI-01 to UI-21)

- **Step 1: Design the overall and detailed Use Case Diagrams.**

- **Mission:**The model has finalized the relationships between the 6 Actors (Administrator, Manager, Farmer, Agronomist, Drone Operator, Technician) and 17 Use Cases.
- **Technique:**Group Use Cases into functional boundaries. Use relationships.<<include>>for the System Login flow (UC-03) and <<extend>>for the emergency SOS command trigger flow (UC-14).

- **Step 2: Build a Data Flow Diagram (DFD)**

- **Mission:**Clarify the data flow from input to storage in the warehouse.
- **Technique:**Decomposition from Context Level (1 central processing) to Level 1 (Divided into major subsystems: Resource Management, Scheduling, Prescription, Operations, Billing) and Level 2 (Detailed device assignment flow)UC-06to see how data interacts with data warehouses.

- **Step 3: Design the Entity-Relationship Diagram (ERD)**

- **Mission:**Build a relational database structure to store system information.
- **Technique:**Define in detail the entities, attributes, primary key (PK), and foreign key (FK) for the core data tables: Users (Users), Land plot (Fields), Mission (MissionsPrescription (Prescriptions), and Flight Log (Operation_Logs).

- **Step 4: Design a high-fidelity UI/UX prototype.**

- **Mission:**Realize the list of 21 screens (UI-01 arrive UI-21) has been defined in the RTM matrix.
- **Technique:**Design a web interface for the management center (statistical dashboard, visual monitoring map) and a mobile interface for field use (large buttons, optimized high contrast for outdoor use, offline map viewing screen). Set up a simulated interactive flow (click-through) for a visual demonstration before the council.

17. AI Audit Log

The log records the interaction between the team and the AI tools (Claude, ChatGPT, Gemini) during Phase 2 – Requirements Engineering (W4), according to two DTC components: Decomposition and Pattern Recognition. Each entry shows a full contextual prompt, a summary of the AI output, the acceptance/modification/rejection decision with reasons, and the corresponding verification step.

STT	Week / Phase	DTC	AI Tool	Prompt	AI Output (summary)	Decision & Refinement	Verification
1	W4 / Phase 2 — Requirements Engineering	Decomposition	ChatGPT	For an "Agriculture UAV" system managing agricultural spraying services using drones with 6 actors (Farmer, Manager, Agronomist, Drone Operator, Technician, Administrator), decompose the business requirement for "account management and centralized permissions" into specific Functional Requirements following the pattern ID - Requirement - Priority - Input - Business Rule - Output.	AI suggests 5 FRs related to account management: account creation, verification, permission management, account locking, and password reset.	Combine the two FRs (account lockout + password reset) into FR-01 because they share the same Admin account management workflow; separate FR-03 Login & Security Authentication because it has its own encryption/JWT business rule; remove the proposed AI FR on "subscription/billing management" because it falls outside the D1 scope (there is no subscription billing module yet).	Compare FR-01 and FR-03 with the RBAC 6-role model defined in Report 1 (Section 4), ensuring that no roles are missing from the authorization flow.

2	W4 / Phase 2 — Requirements Engineering	Pattern	Claude	List common Non-Functional Requirement patterns (according to ISO/IEC 25010) needed for an IoT device operation management system with high legal traceability requirements (audit trail for chemical spraying operations), applicable to agricultural UAV systems.	AI proposes 8 NFR groups based on the ISO 25010 standard (Performance, Security, Usability, Reliability, Maintainability, Portability, Compatibility, Auditability) along with suggested measurement metrics.	Accept the Auditability group as NFR-09 (permanently logs user ID/IP/action) because it directly matches legal requirements; exclude the Portability and Compatibility groups because the project only deploys on one fixed platform; add separate NFR-05/NFR-06 regarding outdoor usability (contrast ratio, instructional videos) which AI does not address due to the specific nature of Drone Operator operation in the field.	Check each NFR for specific measurable data (figures, %, seconds) according to the SRS Quality Checklist (Section 15). Generic NFRs are not measurable.
3	W4 / Phase 2 — Requirements Engineering	Decomposition	Gemini	From the "Agronomist assesses request and creates electronic prescriptions" process of the Agriculture UAV system, break it down into specific Business Rules (BRs) related to controlling prohibited chemicals and tracking approvals.	AI proposes four business rules: checking for banned chemicals, limiting dosages, requiring digital signatures, and logging approvals.	Accept the rule for checking banned chemicals (Banned Chemical Blacklist) and the rule for saving logs to System_Audit_Log (BR-20); merge the "digital signature" and "dosage limit" rules into BR-19 as they both belong to the prescription approval step, avoiding overlap with UC Approve Prescription.	Compare BR-19 and BR-20 with NFR-09 (Auditability) to ensure that business rules and non-functional requirements do not conflict regarding logging behavior.

4	W4 / Phase 2 — Requirements Engineering	Pattern	ChatGPT	Compare two common use case specification methods (fully-dressed use cases according to Cockburn vs. abbreviated table format) to select the appropriate format for the Detailed Use Case Description section of the SRS ITA301 course, which includes approximately 14 use cases.	AI recommends using fully-dressed use cases for all 14 UCs because they are "more complete and professional".	⚠ Recognizing that the recommendation to uniformly apply full-dressing to all 14 UCs is impractical given the D2 timeframe (risk of delays — risk R12 in Report 1); only apply full-dressing to exceptional multi-threaded core UCs (Approve Prescription, Schedule Mission), and use a shortened version for the rest to ensure progress.	Assess the complexity (number of alternative flows) of each UC before deciding on the format, rather than applying AI-driven methods mechanically.
5	W4 / Phase 2 — Requirements Engineering	Decomposition	Claude	The Agriculture UAV system needs to store system audit log data. Suggest the necessary data attributes for this entity to ensure it supports legal tracing and incident investigation.	AI suggested the System_Audit_Logs entity with 6 attributes: audit_id, user_id, action_type, timestamp, ip_address, and description.	Accept all 6 attributes because they match NFR-09 (user ID, timestamp, IP, action details); add "target_entity" which AI did not mention, necessary to differentiate logs of different entities when the Administrator performs error lookup (FR-02).	Compare the System_Audit_Logs entity (Section 12.15) with FR-02 and BR-20 to ensure that sufficient data fields are present to serve the specified function.
6	W4 / Phase 2 — Requirements Engineering	Pattern	Gemini	Propose a Quality Checklist (SRS Quality Checklist) based on IEEE 830 criteria for self-reviewing the Agriculture UAV system's SRS documentation before submitting Deliverable 2.	AI đưa checklist 9 tiêu chí IEEE 830 (Correct, Unambiguous, Complete, Consistent, Ranked, Verifiable, Modifiable,	Accept all 9 criteria as the framework for the SRS Quality Checklist (Section 15); the team will independently supplement the actual Pass/Fail assessments and notes for each criterion based on the completed	Each member independently checks one criterion across the entire SRS document before the final Pass/Fail results are

					Traceable, Understan dable).	FR/NFR/BR content, without copying the criterion descriptions verbatim from the AI.	compiled.
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